TECHNICAL METHODOLOGY REPORT: EVALUATING THE IMPACT OF COMMERCIAL FERTILIZER BRANDS ON MAIZE FIELDS USING ONE WAY ANOVA

1. Introduction and Experimental Design

This report details the statistical procedures used to evaluate the efficacy of five distinct commercial fertilizer brands (Treatments A, B, C, D, and E) on agricultural output. The primary objective is to determine if the variance in crop yield is statistically significant across the treatments.

1. Dependent Variable: Maize Yield (continuous; measured in tons per hectare).
2. Independent Variable (Treatment): Fertilizer Brand (categorical; 5 levels).
3. Design Specifications: The experiment utilizes an unbalanced design, with the number of observations per treatment ranging from n=2 to n=4.
4. Significance Level: All statistical tests in this pipeline are evaluated at a 5% level of significance ($\alpha = 0.05$)

2. Assumption Testing Phase I: Normality of Residuals

A fundamental assumption of the One-Way ANOVA is that the residuals of the model are approximately normally distributed. To ensure robustness and guard against the limitations of any single algorithm, normality was evaluated using a comprehensive suite of both statistical tests and visual diagnostics.

2.1 Statistical Tests for Normality

The null hypothesis (H_0) for all normality tests states that the data follows a normal distribution. If $p > 0.05$, we fail to reject H_0 .

1. Shapiro-Wilk Test: Yielded a p-value of 0.9364. (Interpretation: Since $p > 0.05$, this indicates the residuals are normally distributed. This test is highly sensitive and reliable for small sample sizes.)
2. Kolmogorov-Smirnov Test: Yielded a p-value of 0.9912.
3. Lilliefors Test: Yielded a p-value of 0.9336.
4. Anderson-Darling Test: Yielded a p-value of 0.8767.
5. Cramér-von Mises Test: Yielded a p-value of 0.8989.
6. Pearson Chi-Square Test: Yielded a p-value of 0.6671
7. D'Agostino Test: Yielded a p-value of 0.82.

Normality Conclusion: The consensus across the algorithmic tests indicates that the assumption of normality has been met

2.2 Visual Diagnostics for Normality

Histogram of Standardized Residuals: Visual inspection of the distribution shows a symmetric bell-shape, corroborating the statistical findings.

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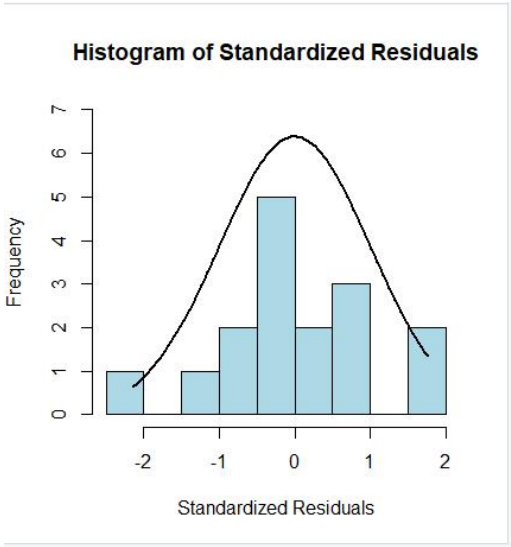
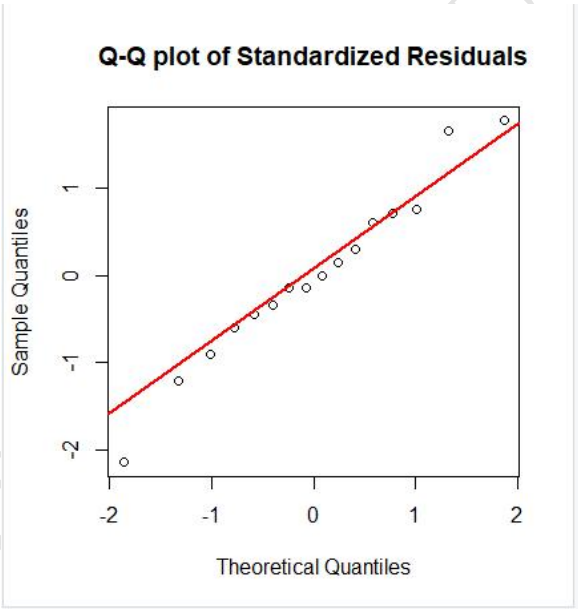


Quartile-Quartile (Q-Q) Plot: The plotted data points adhere closely to the theoretical diagonal reference line, indicating that the quantiles of the residuals match the quantiles of a standard normal distribution.



Kernel Density Plot: The smoothed density curve aligns with the expected Gaussian shape without severe tail deviations.

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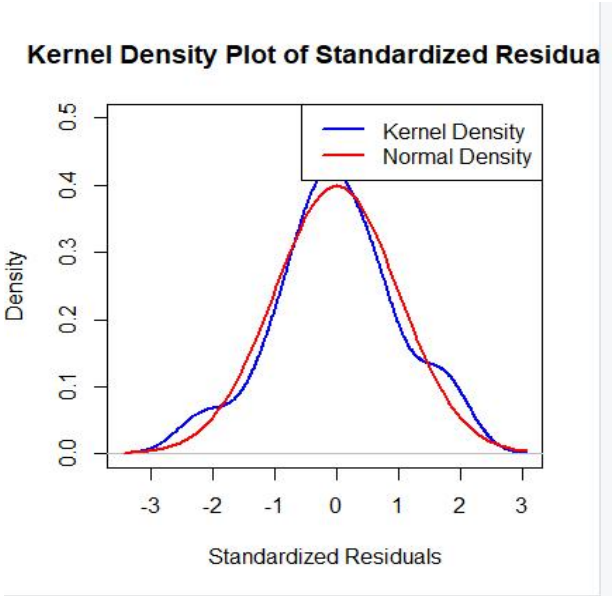

3. Assumption Testing Phase II: Homogeneity of Variance
ANOVA requires the variance among the different treatment groups to be roughly equal (homoscedasticity). The null hypothesis (H_0) states that variances are equal across all groups.

3.1 Statistical Tests for Homogeneity

Bartlett's Test: Yielded a p-value of 0.1126. *(Bartlett's is highly sensitive to deviations from normality).*

Fligner-Killeen Test: Yielded a p-value of 0.3363. *(Fligner-Killeen is a non-parametric test and is highly robust against outliers).*

Homogeneity Conclusion: With p-values greater than 0.05, we fail to reject the null hypothesis. The assumption of equal variances is met, validating the use of a standard One-Way ANOVA.

3.2 Visual Diagnostic

Residuals vs. Fitted Plot: The spread of residuals remains relatively constant across all fitted values, with no obvious funneling patterns, visually confirming homoscedasticity.

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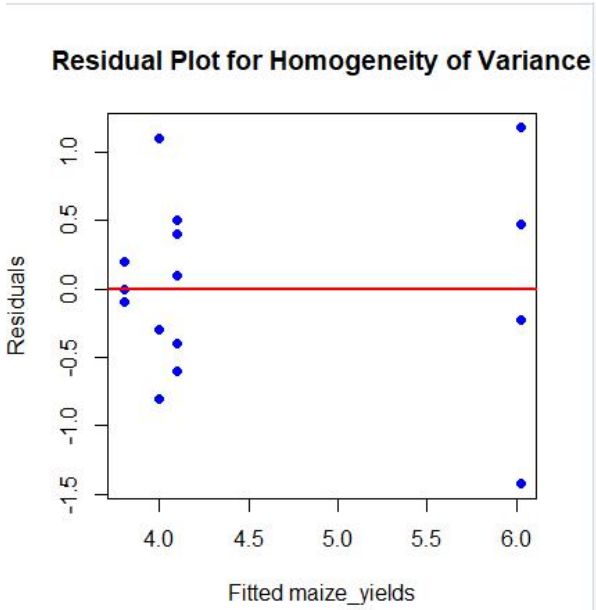

4. Analysis of Variance (ANOVA)
Having satisfied the core assumptions, a One-Way ANOVA was executed to test for differences in mean maize yields across the five fertilizer brands. The model inherently adjusted for the unbalanced sample sizes.

H_0: $\mu_A = \mu_B = \mu_C = \mu_D = \mu_E$ (All fertilizer brands yield the same mean output).

H_A: At least one treatment mean is significantly different.

Results:
The ANOVA yielded an F-statistic of 5.315 with a corresponding p-value of 0.0124. Because $p < 0.05$, we reject the null hypothesis. There is a statistically significant difference in maize yield based on the fertilizer brand applied.

5. Post-Hoc Analysis
Because the ANOVA test was significant, post-hoc procedures were required to determine exactly which specific fertilizer brands differ from one another. To rigorously control for Type I errors (false positives) that arise from multiple comparisons, several correction methods were applied:

5.1 Family-Wise Error Rate (FWER) Controls
These methods strictly control the probability of making even one false discovery across all pairwise comparisons:

Tukey Honestly Significant Difference (HSD): Identified significant mean differences between Brand D and every other Brands

5.2 Unadjusted Comparisons

Least Significant Difference (LSD): While LSD identified differences between Brand D and every other Brands , these results are interpreted with caution as LSD does not correct for multiple comparisons and is prone to inflated Type I error rates.

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5. Final Conclusion & Recommendation

Based on the comprehensive statistical evaluation, commercial fertilizer brand D has a statistically significant impact on maize yield. Specifically, post-hoc analyses confirm that Brand D produces a significantly higher yield per hectare compared to Brands A, B, C. and E

For agricultural operations aiming to maximize crop tonnage, the application of Brand D is the statistically optimal recommendation.

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